Supplementary Information for

A Paired, Floor-Guarded Evaluation Protocol for INT8 and FP8 Object Detectors under Image Corruptions

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S1 Purpose, scope, and evidence hierarchy

This supplement documents the estimands, diagnostic summaries, sensitivity analyses, and artifact contract supporting the main article. It does not add a new population-level claim. The evidence hierarchy used throughout the paper is: (i) the 144-cell four-dataset YOLO landscape for exploratory factorial characterization; (ii) the selection-disjoint VOC/KITTI final holdout as the strongest split-independent conditional evidence; (iii) targeted realization, training, calibration, class-universe, codec, weighting, and runtime sensitivities; and (iv) RT-DETR and RetinaNet experiments as recipe-portability stress cases.

The labels *INT8* and *FP8* denote the complete recorded TensorRT treatments—quantizer placement, Q/DQ coverage, calibration or scale construction, graph rewriting, mixed-precision fallbacks, engine building, and frozen decoding—rather than datatype effects in isolation. All quantities are reported in AP points unless stated otherwise.

S2 Atomic paired design and estimands

For a fixed dataset d, checkpoint m, endpoint b, and corruption condition c, let J95 denote the deterministic JPEG-quality-95 matched-clean control. The explicit J95 label avoids confusion with AP at IoU 0.95. For quantized treatment q, define the FP32-referenced quantization gap and the clean-adjusted excess gap as

$$Q_{d,m,q,b,c} = \mathrm{AP}_{d,m,\mathrm{FP32},b,c} - \mathrm{AP}_{d,m,q,b,c}, \tag{S1}$$

$$E_{d,m,q,b,c} = Q_{d,m,q,b,c} - Q_{d,m,q,b,\mathrm{J95}}. \tag{S2}$$

A positive E means that corruption widens the treatment–FP32 discrepancy relative to matched clean; a negative E means that the discrepancy contracts. Neither sign is, by itself, evidence of retained absolute accuracy.

The direct treatment interaction is computed inside each common-image resample:

$$\Delta E_{d,m,b,c} = E_{d,m,\mathrm{INT8},b,c} - E_{d,m,\mathrm{FP8},b,c} \tag{S3}$$

$$= \big(\mathrm{AP}_{\mathrm{FP8},c} - \mathrm{AP}_{\mathrm{INT8},c}\big) - \big(\mathrm{AP}_{\mathrm{FP8},\mathrm{J95}} - \mathrm{AP}_{\mathrm{INT8},\mathrm{J95}}\big). \tag{S4}$$

The FP32 term cancels algebraically. Consequently, ΔE asks whether the FP8–INT8 discrepancy changes under corruption after removing the discrepancy already present on matched clean inputs. A positive value indicates expansion; a negative value indicates contraction. This is a descriptive two-by-two interaction contrast, not a causal difference-in-differences estimator.

For area endpoints, the within-treatment size contrast is

$$\Psi_{d,m,q,c} = E_{d,m,q,S,c} - E_{d,m,q,L,c}, \tag{S5}$$

with direct size interaction $\Delta\Psi = \Psi_{\mathrm{INT8}} - \Psi_{\mathrm{FP8}}$. TT100K uses its native height-based small-like and large-like endpoints, which are reported separately and never pooled numerically with the COCO/VOC/KITTI area endpoints.

The signed clean-to-corrupted AP loss used by the absolute-accuracy guardrail is

$$D_{d,m,p,b,c} = \mathrm{AP}_{d,m,p,b,\mathrm{J95}} - \mathrm{AP}_{d,m,p,b,c}. \tag{S6}$$

Thus $D > 0$ is an AP loss and $D < 0$ is an AP gain; despite its role in the absolute-accuracy guardrail, D is not an absolute value. The direct interaction can therefore also be written as

$$\Delta E_{d,m,b,c} = D_{d,m,\mathrm{INT8},b,c} - D_{d,m,\mathrm{FP8},b,c}. \tag{S7}$$

The protocol is therefore *floor-guarded*: D, corrupted AP, and low-AP inventories are reported beside relative interactions. It is not a formal statistical correction for censoring or an AP floor.

As a secondary scale sensitivity, relative retention normalizes each corrupted arm by that treatment's own matched-clean AP:

$$R_{d,m,q,b,c} = 100\frac{\mathrm{AP}_{d,m,q,b,c}}{\mathrm{AP}_{d,m,q,b,\mathrm{J95}}}, \tag{S8}$$

$$\Delta R_{d,m,b,c} = R_{d,m,\mathrm{FP8},b,c} - R_{d,m,\mathrm{INT8},b,c}. \tag{S9}$$

ΔR is reported in percentage points (pp); positive values mean that FP8 retains a larger percentage of its own matched-clean AP. Because this ratio changes the scale and effective weighting induced by clean AP, it is a sensitivity estimand rather than a replacement for the AP-point interaction ΔE.

The image is the resampling unit. Each direct comparison reuses one deterministic schedule of 2,000 image-index bootstrap draws across all four INT8/FP8 by clean/corrupted arms. Percentile intervals quantify finite-evaluation-set uncertainty conditional on the frozen datasets, checkpoints, calibration lists, engines, corruption materializations, and decoder. They do not propagate a population distribution over datasets, training runs, calibration runs, or corruption generators.

S3 Decision-impact diagnostic

The methodological value of clean adjustment can be measured directly. For each of the 144 exploratory cells, the raw corrupted gap satisfies

$$(\mathrm{AP}_{\mathrm{FP8}} - \mathrm{AP}_{\mathrm{INT8}})_c = (\mathrm{AP}_{\mathrm{FP8}} - \mathrm{AP}_{\mathrm{INT8}})_{\mathrm{J95}} + \Delta E. \tag{S10}$$

The raw corrupted comparison favored FP8 in 134 cells, but 56 of those cells had negative ΔE point estimates after subtracting the matched-clean discrepancy (Table S1). Of these 56 point-sign reversals, 17 paired percentile intervals lay entirely below zero and 39 crossed zero; 56 is therefore not a count of statistically resolved discoveries. Thus a raw corrupted leaderboard comparison would conflate clean fidelity with corruption-specific interaction in 38.9% of the finite grid.

Table S1: Decision-impact inventory over the 144 exploratory cells. "Below AP" counts a cell when at least one quantized treatment crosses the stated floor.

Diagnostic	Cells
Raw corrupted FP8–INT8 gap > 0	134/144
Raw corrupted FP8–INT8 gap < 0	10/144
Clean-adjusted $\Delta E > 0$	78/144
Clean-adjusted $\Delta E < 0$	66/144
Raw-positive but adjusted-negative	56/144
At least one format below 10 AP	18/144
At least one format below 5 AP	5/144

Table S2: Sensitivity of the 56 positive-raw-gap to negative-ΔE point-sign reversals to minimum interaction magnitude. Intervals are paired image-bootstrap percentile intervals and are not multiplicity-adjusted.

Point-effect filter (AP)	Reversals	Interval below zero	Interval crosses zero
$\lvert\Delta E\rvert \geq 0.00$	56	17	39
$\lvert\Delta E\rvert \geq 0.25$	34	17	17
$\lvert\Delta E\rvert \geq 0.50$	20	14	6
$\lvert\Delta E\rvert \geq 1.00$	11	10	1

S4 Quantized graph coverage

Table S3 summarizes hash-bound ONNX graph coverage. These counts characterize inserted Q/DQ structure and zero-point datatype; they are not a claim that every TensorRT kernel executes at the nominal precision. That stronger claim would require per-layer engine inspection or profiling, which is outside the recorded evidence package.

Table S3: Graph-level Q/DQ coverage across the evaluated detector families. Ranges reflect the three dataset-specific graphs for each row.

Family	Model	Format	Graphs	Q	Weight Q	Act. Q	Direct Q/DQ nodes	Zero point
YOLO11	YOLO11m	INT8	3	437–441	113	324–328	352–355/405	INT8
YOLO11	YOLO11m	FP8	3	213–219	101–104	112–115	133–136/405	FLOAT8E4M3FN
YOLO11	YOLO11n	INT8	3	325–341	85–88	240–253	263–275/320	INT8
YOLO11	YOLO11n	FP8	3	152–158	69–72	83–86	99–102/320	FLOAT8E4M3FN
YOLO11	YOLO11x	INT8	3	682–690	174	508–516	546–552/617	INT8
YOLO11	YOLO11x	FP8	3	346–352	161–164	185–188	215–218/617	FLOAT8E4M3FN
RT-DETR	RT-DETR-L	INT8	3	534–538	197–199	335–339	351–353/963	INT8
RT-DETR	RT-DETR-L	FP8	3	398–401	167–169	231–232	251–253/963	FLOAT8E4M3FN
RetinaNet	R50-FPN-v2	INT8	3	217	111	106	139/1076	INT8
RetinaNet	R50-FPN-v2	FP8	3	214	110	104	137/1076	FLOAT8E4M3FN

S5 Exploratory interaction summaries

Table S4 reports the four dataset margins, while Table S5 reports capacity, corruption, and ordinal-severity margins. Cell-wise percentile signs are descriptive inventories rather than multiplicity-adjusted discoveries. The balanced macro is the arithmetic mean of the finite, defined 144-cell grid; it is not a deployment-population weighted effect.

Table S4: Dataset-level direct interaction summaries. TT100K's $\Delta\Psi$ uses the height endpoint; the other rows use the area endpoint.

Dataset	Cells	Mean ΔE (AP pt)	95% percentile sign (+/-/cross)	Mean $\Delta\Psi$ (AP pt)
COCO	36	-0.19	6 / 10 / 20	-0.12
VOC	36	-0.04	12 / 5 / 19	-0.44
KITTI	36	-0.46	5 / 11 / 20	-0.95
TT100K	36	+0.61	2 / 0 / 34	-1.38

Table S5: Factor-level heterogeneity in the exploratory direct ledger.

Factor	Level	Cells	Mean ΔE	Range ΔE	95% percentile sign (+/-/cross)
Checkpoint rung	YOLO11n	48	+0.07	-2.71 to +3.10	11 / 7 / 30
Checkpoint rung	YOLO11m	48	-0.11	-4.28 to +2.82	9 / 10 / 29
Checkpoint rung	YOLO11x	48	-0.02	-4.23 to +2.09	5 / 9 / 34
Corruption	Gaussian noise	36	-0.03	-3.66 to +2.82	9 / 11 / 16
Corruption	Motion blur	36	-0.24	-4.28 to +1.48	4 / 8 / 24
Corruption	Fog	36	+0.39	-0.56 to +3.10	6 / 0 / 30
Corruption	JPEG	36	-0.20	-4.23 to +2.01	6 / 7 / 23
Severity	1	48	+0.28	-1.03 to +2.82	8 / 2 / 38
Severity	3	48	+0.16	-3.37 to +2.09	10 / 8 / 30
Severity	5	48	-0.51	-4.28 to +3.10	7 / 16 / 25

S6 Relative-retention scale sensitivity

Table S6 compares the AP-point interaction ΔE with the relative-retention interaction ΔR. The exploratory balanced means are -0.02 AP points and $+0.75$ pp, respectively, and 18 of 144 cell point estimates differ in sign. The untouched holdouts have means of -0.55 AP points and -0.12 pp, with seven of 72 point-sign disagreements. These differences are expected because the ratio estimand normalizes each treatment by a different matched-clean denominator; they reinforce reporting absolute corrupted AP and ΔE rather than substituting one normalized score.

Table S6: Relative-retention sensitivity. Mean ΔE is in AP points and mean ΔR is in percentage points (pp). Sign differences compare finite cell point estimates, not interval decisions.

Scope	Cells	Mean ΔE (AP)	Mean ΔR (pp)	Sign differences
Exploratory grid	144	-0.02	+0.75	18
Exploratory: COCO	36	-0.19	+0.16	1
Exploratory: VOC	36	-0.04	+0.35	6
Exploratory: KITTI	36	-0.46	+0.49	6
Exploratory: TT100K	36	+0.61	+1.98	5
Final holdouts	72	-0.55	-0.12	7
Final: VOC	36	-0.43	-0.34	3
Final: KITTI	36	-0.67	+0.10	4

Table S7: All 18 exploratory cells for which the point estimates of ΔE and ΔR have different signs. The seven holdout disagreements are summarized in Table S6.

Dataset	Model	Corruption	Severity	ΔE (AP)	ΔR (pp)
COCO	YOLO11n	Gaussian noise	5	-0.31	+1.23
VOC	YOLO11m	Fog	3	-0.02	+0.01
VOC	YOLO11m	Gaussian noise	3	-0.23	+0.14
VOC	YOLO11m	Motion blur	3	-0.34	+1.13
VOC	YOLO11m	Motion blur	5	-1.09	+0.67
VOC	YOLO11n	Gaussian noise	5	-0.11	+0.66
VOC	YOLO11n	JPEG	5	-0.10	+0.17
KITTI	YOLO11m	Gaussian noise	3	-0.33	+3.01
KITTI	YOLO11m	Gaussian noise	5	-3.66	+0.23
KITTI	YOLO11m	JPEG	5	-0.16	+0.06
KITTI	YOLO11n	Gaussian noise	5	-1.13	+1.21
KITTI	YOLO11n	Motion blur	5	-1.70	+0.36
KITTI	YOLO11x	Motion blur	5	-1.18	+0.15
TT100K	YOLO11m	Gaussian noise	5	-1.21	+0.03
TT100K	YOLO11m	Motion blur	5	-0.26	+1.95
TT100K	YOLO11n	Gaussian noise	5	-0.72	+0.57
TT100K	YOLO11n	Motion blur	5	-0.03	+3.78
TT100K	YOLO11x	Motion blur	5	-0.03	+0.07

S7 Size-specific absolute-accuracy guardrail

Table S8 binds relative size interactions to absolute matched-clean AP, corrupted AP, and clean-to-corrupted loss. The AP< 5 inventory makes explicit when an apparent contraction occurs close to a task failure floor. COCO, VOC, and KITTI use COCO-style area endpoints; TT100K uses its original-box-height endpoints.

Table S8: Size-specific absolute-accuracy guardrail. Values are balanced means across 36 model–corruption–severity cells per row.

Dataset	Format	Endpoint	Cells	Clean AP	Corrupted AP	Mean D (AP pt)	AP¡5
COCO	INT8	coco-area:small	36	28.05	18.52	+9.53	5
COCO	INT8	coco-area:large	36	61.75	49.36	+12.39	0
COCO	FP8	coco-area:small	36	29.01	19.37	+9.64	4
COCO	FP8	coco-area:large	36	63.33	50.94	+12.38	0
KITTI	INT8	coco-area:small	36	50.90	34.18	+16.72	6
KITTI	INT8	coco-area:large	36	74.89	54.12	+20.77	2
KITTI	FP8	coco-area:small	36	54.02	36.52	+17.50	6
KITTI	FP8	coco-area:large	36	76.25	55.64	+20.61	2
TT100K	INT8	original-height:XS	36	12.02	6.42	+5.60	20
TT100K	INT8	original-height:S	36	38.03	25.46	+12.57	8
TT100K	INT8	original-height:L	36	58.50	43.67	+14.83	1
TT100K	INT8	original-height:XL	36	65.22	51.82	+13.40	0
TT100K	FP8	original-height:XS	36	14.00	6.93	+7.06	21
TT100K	FP8	original-height:S	36	40.15	27.11	+13.04	8
TT100K	FP8	original-height:L	36	58.52	44.97	+13.55	0
TT100K	FP8	original-height:XL	36	66.42	53.50	+12.93	0
VOC	INT8	coco-area:small	36	28.12	19.98	+8.14	6
VOC	INT8	coco-area:large	36	71.96	54.99	+16.97	0
VOC	FP8	coco-area:small	36	28.97	20.51	+8.46	6
VOC	FP8	coco-area:large	36	72.97	56.12	+16.85	0

For TT100K, Table S9 reports the combined small-like and large-like endpoints used by the height interaction itself. These endpoints are evaluated directly; they are not arithmetic averages of the component XS/S or L/XL AP values, and the medium-height band is excluded by the frozen endpoint definition.

Table S9: TT100K absolute-accuracy guardrail for the combined original-height endpoints used in $\Delta\Psi$. D is the signed clean-to-corrupted AP loss.

Treatment	Original-height endpoint	Cells	Clean AP	Corrupted AP	Mean D (AP)	AP< 5	AP< 10
INT8	small-like $[0,24)$	36	37.74	25.25	+12.49	8	9
INT8	large-like $[48,\infty)$	36	55.61	41.57	+14.04	1	2
FP8	small-like $[0,24)$	36	39.92	26.91	+13.01	8	9
FP8	large-like $[48,\infty)$	36	56.29	43.11	+13.18	0	2

S8 Codec-control sensitivity

The primary estimand uses deterministic JPEG-95 matched-clean inputs so that both clean and corrupted arms share the terminal materialization codec. Table S10 compares original-source clean AP with JPEG-95 clean AP. This point sensitivity quantifies the control choice; it is not a bootstrap interval and does not replace the codec-matched estimand.

Table S10: Original-source clean versus deterministic JPEG-95 clean AP. The last column is JPEG-95 minus original-source AP, all on the 0–100 AP-point scale.

Dataset	Model	Precision	Original clean AP (points)	JPEG-95 clean AP (points)	Codec minus original (AP points)
COCO	YOLO11n	FP32	38.73	38.70	-0.035
COCO	YOLO11n	INT8	36.64	36.65	+0.014
COCO	YOLO11n	FP8	37.74	37.81	+0.072
COCO	YOLO11m	FP32	50.68	50.57	-0.113
COCO	YOLO11m	INT8	48.85	48.82	-0.025
COCO	YOLO11m	FP8	50.08	50.05	-0.028
COCO	YOLO11x	FP32	53.81	53.69	-0.118
COCO	YOLO11x	INT8	52.75	52.56	-0.197
COCO	YOLO11x	FP8	53.52	53.47	-0.056
VOC	YOLO11n	FP32	62.54	62.56	+0.013
VOC	YOLO11n	INT8	61.42	61.51	+0.090
VOC	YOLO11n	FP8	62.06	62.19	+0.125
VOC	YOLO11m	FP32	68.53	68.56	+0.036
VOC	YOLO11m	INT8	66.75	66.66	-0.089
VOC	YOLO11m	FP8	68.31	68.36	+0.044
VOC	YOLO11x	FP32	70.09	70.05	-0.032
VOC	YOLO11x	INT8	69.07	69.03	-0.047
VOC	YOLO11x	FP8	70.00	69.94	-0.060
KITTI	YOLO11n	FP32	64.79	64.84	+0.049
KITTI	YOLO11n	INT8	61.52	61.69	+0.170
KITTI	YOLO11n	FP8	63.68	63.83	+0.147
KITTI	YOLO11m	FP32	73.82	73.87	+0.049
KITTI	YOLO11m	INT8	69.22	69.34	+0.124
KITTI	YOLO11m	FP8	73.32	73.31	-0.016
KITTI	YOLO11x	FP32	72.93	73.06	+0.127
KITTI	YOLO11x	INT8	70.87	70.82	-0.053
KITTI	YOLO11x	FP8	72.20	72.26	+0.061
TT100K	YOLO11n	FP32	28.78	28.76	-0.013
TT100K	YOLO11n	INT8	25.10	25.29	+0.191
TT100K	YOLO11n	FP8	27.12	26.41	-0.709
TT100K	YOLO11m	FP32	56.57	56.55	-0.014
TT100K	YOLO11m	INT8	53.05	52.97	-0.074
TT100K	YOLO11m	FP8	54.75	54.40	-0.350
TT100K	YOLO11x	FP32	57.63	57.66	+0.034
TT100K	YOLO11x	INT8	55.50	56.02	+0.521
TT100K	YOLO11x	FP8	56.76	56.10	-0.667

S9 Weighting, omission, and runtime sensitivities

Alternative weighting changes the descriptive target, not merely the standard error. Table S11 therefore reports equal-cell, image-count, and object-count means together with leave-one-dataset and leave-one-corruption summaries. Sign changes near zero expose conditional heterogeneity; they do not identify a causal role for the omitted factor.

Table S11: Weighting and leave-one-factor-out sensitivity of the 144-cell ledger.

Analysis	Level	Cells	Mean ΔE (AP pt)	SD / IQR (AP pt)
Cell distribution	all cells	144	-0.02	1.20 / [-0.39, +0.56]
Weighting	equal cell	144	-0.02	–
Weighting	image weighted	144	+0.00	–
Weighting	object weighted	144	-0.09	–
Leave-one-dataset-out	COCO	108	+0.03	–
Leave-one-dataset-out	KITTI	108	+0.13	–
Leave-one-dataset-out	TT100K	108	-0.23	–
Leave-one-dataset-out	VOC	108	-0.01	–
Leave-one-corruption-out	Fog	108	-0.16	–
Leave-one-corruption-out	Gaussian noise	108	-0.02	–
Leave-one-corruption-out	JPEG	108	+0.04	–
Leave-one-corruption-out	Motion blur	108	+0.05	–

Runtime is reported only for matched engine conditions. Ratios exclude image decoding, preprocessing, host–device transfer, detector decoding, NMS, power, and energy. Table S12 stratifies the engine-only measurements by input size and YOLO11 capacity rather than pooling unlike workloads into one deployment claim.

Table S13 exposes all 12 matched conditions underlying the ratio summaries. Ratios are computed condition by condition; their medians therefore need not equal ratios formed from separately pooled marginal medians.

Table S12: Matched engine-only latency-ratio sensitivity. Entries are median [Q1, Q3] across retained conditions.

Stratum	Level	Conditions	FP32/INT8	FP32/FP8	INT8/FP8
Input size	640	9	3.08 [1.56, 3.08]	3.32 [1.81, 3.33]	1.09 [1.08, 1.16]
Input size	1280	3	2.81 [2.38, 2.84]	2.86 [2.33, 2.91]	1.02 [0.97, 1.02]
Capacity	YOLO11m	4	3.08 [3.01, 3.09]	3.34 [3.21, 3.35]	1.08 [1.06, 1.09]
Capacity	YOLO11n	4	1.55 [1.52, 1.65]	1.80 [1.79, 1.81]	1.17 [1.10, 1.18]
Capacity	YOLO11x	4	3.06 [3.00, 3.09]	3.32 [3.22, 3.33]	1.07 [1.05, 1.08]

Table S13: Condition-level TensorRT engine-only latency and matched ratios for the primary YOLO11 grid.

Dataset	Model	FP32 ms	INT8 ms	FP8 ms	32/INT8	32/FP8	INT8/FP8
COCO	YOLO11m	2.445	0.792	0.734	3.09	3.33	1.08
COCO	YOLO11n	0.543	0.365	0.306	1.49	1.78	1.19
COCO	YOLO11x	5.336	1.753	1.613	3.04	3.31	1.09
VOC	YOLO11m	2.422	0.786	0.725	3.08	3.34	1.08
VOC	YOLO11n	0.550	0.354	0.305	1.56	1.80	1.16
VOC	YOLO11x	5.367	1.709	1.609	3.14	3.33	1.06
KITTI	YOLO11m	2.422	0.785	0.720	3.08	3.36	1.09
KITTI	YOLO11n	0.551	0.358	0.304	1.54	1.81	1.18
KITTI	YOLO11x	5.322	1.730	1.602	3.08	3.32	1.08
TT100K	YOLO11m	5.537	1.970	1.936	2.81	2.86	1.02
TT100K	YOLO11n	1.406	0.723	0.785	1.95	1.79	0.92
TT100K	YOLO11x	12.965	4.516	4.393	2.87	2.95	1.03

S10 Selection-disjoint holdout and corruption-realization sensitivities

The selection-disjoint branch froze disjoint checkpoint-selection and final image lists before training. VOC used 8,218 training, 2,510 selection, and 5,823 final images. For KITTI, the 1,496-image selection partition was reserved first; a deterministic class-covered 20% subset of the remaining 5,985 images formed the 1,197-image final set, and the other 4,788 images formed the training set (Table S14). Calibration identities were drawn only from training data. The final factorial rerun retained 72 equal-weight VOC/KITTI cells and reused a common 2,000-draw schedule within each dataset. Its balanced direct interaction was -0.55 AP points with percentile interval $[-0.78, -0.29]$; VOC was -0.43 $[-0.54, -0.29]$ and KITTI was -0.67 $[-1.13, -0.20]$. FP8 retained 1.60 more matched-clean AP on average, and the clean advantage was positive in all six dataset–capacity blocks. Nine of 144 corrupted format arms were below 5 AP and 18 were below 10 AP. This branch removes evaluation-after-selection reuse for these two datasets, but it does not convert the broader exploratory grid into a prospective confirmatory study.

Table S14: Selection-disjoint partition counts. The KITTI final set is a class-covered subset of the post-selection remainder, not 20% of the complete 7,481-image labeled universe.

Dataset	Partition	Images	Objects	Observed/declared classes
VOC	Train	8,218	19,910	20/20
VOC	Selection	2,510	6,307	20/20
VOC	Final	5,823	13,841	20/20
KITTI	Train	4,788	25,847	8/8
KITTI	Selection	1,496	8,128	8/8
KITTI	Final	1,197	6,595	8/8

The frozen reference ladder was checked on the final partitions before the quantized comparisons. Table S15 reports absolute AP-point gaps; all six blocks satisfied the recorded parity gate.

The corruption-realization sensitivity used three fixed realization seeds over 36 base conditions on class-covering 512-image subsets of VOC, KITTI, and TT100K. Its equal-cell mean ΔE was -0.23 AP points, with nested realization/image percentiles $[-0.95, -0.28, +0.37]$. Mean between-realization variance pooled over stochastic and

Table S15: Selection-disjoint reference-ladder parity. Values are absolute AP-point gaps between the named implementations.

Dataset	Model	Source–ORT	Source–TRT32	TRT32–TRT16	Gate
VOC	YOLO11m	0.0013	0.0024	0.0004	Pass
VOC	YOLO11n	0.0027	0.0052	0.0008	Pass
VOC	YOLO11x	0.0041	0.0042	0.0213	Pass
KITTI	YOLO11m	0.0362	0.0404	0.1084	Pass
KITTI	YOLO11n	0.0221	0.0036	0.0123	Pass
KITTI	YOLO11x	0.0001	0.0126	0.2554	Pass

deterministic conditions was 0.3989 AP-points^2, compared with 0.7446 within-image-bootstrap variance. The more appropriate stochastic-only between-realization mean was 0.5318 AP-points^2; deterministic JPEG has no realization variance by construction and is marked not applicable in Table S17. For $\Delta\Psi$, the mean was -1.04 with nested percentiles $[-3.05, -1.42, +0.10]$. Only three materializations were used, so these are conditional sensitivity summaries rather than population intervals over all possible corruptions. In this separate sensitivity branch, stochastic fields and motion-blur angles were nested across severity; in the exploratory grid, severity remains an ordinal factor conditional on its recorded materialization.

Table S16: Realization-specific direct interactions. SD is the descriptive standard deviation across the three fixed realization labels.

Realization seed	Cells	Mean ΔE (AP)	Mean $\Delta\Psi$ (AP)
202608181	36	-0.12	-1.27
202608182	36	-0.26	-0.83
202608183	36	-0.32	-1.01
Across-seed summary	108	-0.23; SD 0.10	-1.04; SD 0.22

Table S17: Per-corruption realization sensitivity. Variances are in squared AP points. JPEG is deterministic and is excluded from the stochastic-only mean.

Corruption	Mean ΔE (AP)	Between-realization variance	Within-image variance
Gaussian noise	-0.12	0.5960	0.7306
Motion blur	-0.86	0.5931	0.8302
Fog	+0.23	0.4063	0.7114
JPEG	-0.18	n/a (deterministic)	0.7062
Stochastic-only mean	–	0.5318	–

Table S18: Scope and uncertainty target of the four conditional evidence layers. The reported intervals are not interchangeable estimates of one parameter.

Analysis	Dataset/model/severity scope	Cells	Mean ΔE	Interval (AP)	Uncertainty target
Exploratory grid	4; n/m/x; S1/3/5	144	-0.02	[-0.24, +0.15]	Paired image bootstrap
Final holdouts	VOC/KITTI; n/m/x; S1/3/5	72	-0.55	[-0.78, -0.29]	Paired image bootstrap
Realization subset	3 datasets; m; S1/3/5	36 base / 108	-0.23	[-0.95, +0.37]	3-label nested sensitivity
Training/calibration seeds	3 datasets; m; S5	108	-1.11	not estimated	Descriptive 3×3 grid

S11 Training–calibration seed sensitivity

The targeted 3×3 training–calibration grid covers YOLO11m on VOC, KITTI, and TT100K at severity 5. Across 108 cells, the equally weighted mean ΔE was -1.11 AP points. Training-seed marginal means ranged from -1.64 to -0.68, and calibration-seed marginal means from -1.32 to -0.84. The original-seed intersection was -1.21, close to the balanced seed-grid mean. These results do not extend to COCO, other capacity rungs, lower severities, or an unspecified population of training and calibration seeds.

Table S19 records the optimizer realization and the checkpoint-selection outcome for each exploratory transfer or untouched-holdout training run for which training was performed in this study. COCO uses an official pretrained checkpoint and is therefore absent. Selection AP is not a final-evaluation result.

Table S19: Training realizations and selected best epochs. Effective weight decay records the realized batch/accumulation scaling. AP values are on the 0–100 scale and refer only to the checkpoint-selection partition.

Branch	Dataset	Model	Batch	Optimizer	Base LR	Mom./β_1	Effective WD	Best epoch	Selection $AP_{50:95}$
Expl.	VOC	YOLO11n	52	SGD-momentum	0.01	0.9	0.00040625	99	63.653
Expl.	VOC	YOLO11m	14	SGD-momentum	0.01	0.9	0.000546875	65	69.593
Expl.	VOC	YOLO11x	6	SGD-momentum	0.01	0.9	0.000515625	67	71.228
Expl.	KITTI	YOLO11n	55	AdamW	0.000833	0.9	0.0004296875	100	65.203
Expl.	KITTI	YOLO11m	14	AdamW	0.000833	0.9	0.000546875	91	73.850
Expl.	KITTI	YOLO11x	6	AdamW	0.000833	0.9	0.000515625	100	73.012
Expl.	TT100K	YOLO11n	11	AdamW	4.4e-05	0.9	0.000515625	81	7.197
Expl.	TT100K	YOLO11m	3	AdamW	4.4e-05	0.9	0.0004921875	74	26.746
Expl.	TT100K	YOLO11x	1	AdamW	4.4e-05	0.9	0.0005	78	26.117
Holdout	VOC	YOLO11n	16	SGD-momentum	0.01	0.9	0.0005	82	57.925
Holdout	VOC	YOLO11m	16	SGD-momentum	0.01	0.9	0.0005	64	63.428
Holdout	VOC	YOLO11x	16	SGD-momentum	0.01	0.9	0.0005	73	63.662
Holdout	KITTI	YOLO11n	16	AdamW	0.000833	0.9	0.0005	100	62.026
Holdout	KITTI	YOLO11m	16	AdamW	0.000833	0.9	0.0005	99	71.211
Holdout	KITTI	YOLO11x	16	AdamW	0.000833	0.9	0.0005	98	71.811

Table S20: Descriptive two-way seed decomposition within each targeted YOLO11m severity-5 block. The last three columns are percentages of the within-block total sum of squares; Train×cal. denotes non-additivity.

Dataset	Corruption	Mean	Train (%)	Cal. (%)	Train×cal. (%)
VOC	Gaussian noise	-1.494	28.6	23.0	48.4
VOC	Motion blur	-1.256	11.3	18.0	70.7
VOC	Fog	+0.018	34.2	36.5	29.3
VOC	JPEG	-2.033	9.3	80.2	10.5
KITTI	Gaussian noise	-3.532	51.5	13.8	34.8
KITTI	Motion blur	-4.350	57.1	37.4	5.5
KITTI	Fog	+1.608	23.9	24.8	51.3
KITTI	JPEG	+0.120	38.4	54.6	7.0
TT100K	Gaussian noise	-1.381	8.7	22.5	68.8
TT100K	Motion blur	-0.998	12.6	5.5	82.0
TT100K	Fog	+0.769	11.5	27.5	61.0
TT100K	JPEG	-0.787	90.9	0.5	8.5

S12 Fixed-class-universe bootstrap sensitivity

For the targeted TT100K YOLO11m/motion-blur/severity-5 stress cell, a positive-weight Bayesian image bootstrap retained the complete category universe across all 3,067 images. Each draw assigned independent unit-rate exponential image weights, normalized the 3,067 weights to mean one, applied them to ground-truth positives and true/false-positive contributions, and reused the same vector across all four arms. With 10,000 draws, the ΔE percentiles were $[-1.19, -0.17, +0.88]$, compared with $[-1.79, -0.20, +1.34]$ under the ordinary 2,000-draw image bootstrap. The height-based $\Delta\Psi$ percentiles were $[-5.69, -4.08, -2.48]$ versus $[-7.22, -4.48, -1.81]$. The median sign and zero-crossing status of ΔE, and the negative-interval status of $\Delta\Psi$, were unchanged. This targeted cell does not establish class-composition invariance for the full TT100K grid.

S13 Architecture-portability stress cases

The RT-DETR-L and RetinaNet-R50-FPN-v2 extension tests whether the frozen quantization recipes transfer beyond the YOLO family. It is interpreted in a fixed sequence: matched-clean fidelity first, absolute corrupted AP second, and the corruption interaction third.

Table S21: Matched JPEG-95 clean AP and clean quantization gaps for the architecture-portability stress cases.

Dataset	Family	FP32 AP	INT8 AP	FP8 AP	$Q_{\mathrm{J95}}^{\mathrm{INT8}}$	$Q_{\mathrm{J95}}^{\mathrm{FP8}}$
VOC	RT-DETR-L	70.44	1.14	70.14	69.30	0.30
VOC	RetinaNet-R50-FPN-v2	57.22	33.69	56.17	23.53	1.05
KITTI	RT-DETR-L	12.29	3.32	12.02	8.97	0.27
KITTI	RetinaNet-R50-FPN-v2	50.43	7.13	49.27	43.30	1.15
TT100K	RT-DETR-L	43.17	5.46	42.13	37.71	1.04
TT100K	RetinaNet-R50-FPN-v2	3.37	2.60	3.38	0.77	-0.01

Table S22: Canonical cross-family interaction summaries under $E = Q_c - Q_{\mathrm{J95}}$. Negative INT8 means largely reflect contraction of a large clean gap near low corrupted AP.

Family	Format	Mean Q_{J95}	Mean E	E range	Mean corrupted AP	AP$<$ 5
RT-DETR-L	INT8	38.66	-6.81	-54.39–+1.22	1.97	34/36
RT-DETR-L	FP8	0.54	-0.02	-1.06–+2.01	33.30	4/36
RetinaNet-R50-FPN-v2	INT8	22.53	-6.21	-38.86–+1.14	11.34	16/36
RetinaNet-R50-FPN-v2	FP8	0.73	+0.01	-1.11–+0.64	26.91	12/36

Table S23: Descriptive direct ΔE summaries for the architecture stress cases. These values are not evidence of intrinsic datatype superiority.

Family	Cells	Mean ΔE	Minimum	Maximum
RT-DETR-L	36	-6.79	-54.33	+1.51
RetinaNet-R50-FPN-v2	36	-6.23	-37.75	+1.16

Direct contrasts are computed from unrounded cell values; consequently, displayed rounded E_{INT8} and E_{FP8} components need not subtract to the displayed direct ΔE exactly.

FP8 remained within 1.15 matched-clean AP points of the recorded high-precision TensorRT reference in all six dataset–family blocks. By contrast, the transferred INT8 recipe produced mean clean gaps of 38.66 AP for RT-DETR-L and 22.53 for RetinaNet. Mean corrupted INT8 AP was 1.97 and 11.34, respectively. The supported conclusion is therefore a failure of recipe portability under these recorded treatments, not an intrinsic limitation of INT8 or a universal advantage of FP8.

Table S24: Per-dataset matched engine-only runtime for the architecture stress cases. Ratios are formed within each dataset and family.

Family	Dataset	FP32 ms	INT8 ms	FP8 ms	32/INT8	32/FP8
RT-DETR-L	VOC	2.770	1.120	1.203	2.47	2.30
RT-DETR-L	KITTI	2.756	1.121	1.197	2.46	2.30
RT-DETR-L	TT100K	6.814	2.434	2.591	2.80	2.63
RetinaNet-R50-FPN-v2	VOC	5.080	1.113	1.290	4.56	3.94
RetinaNet-R50-FPN-v2	KITTI	4.994	1.085	1.281	4.60	3.90
RetinaNet-R50-FPN-v2	TT100K	14.285	3.613	4.232	3.95	3.38

S14 Evidence-generation and audit boundary

The versioned public evidence archive cited in the main article separates three layers:

- **Metric ledgers and evidence reports** contain the retained summaries and upstream SHA-256 bindings supplied with the study.
- **Generation scripts** deterministically reconstruct the decision-impact diagnostic and the canonical cross-family sign convention from packaged CSV inputs.
- **LaTeX** consumes prevalidated tables and figures; compilation itself does not recompute model inference or independently verify external files that are absent from the compact journal source package.

The external research pipeline validated metric-to-run, input-manifest, ordered-image, prediction, and analysis hashes before generating the packaged artifacts. The manuscript does not claim that the LaTeX engine performs those checks. The validator archived with the public research release checks package-level consistency: row counts, formula signs, required files, abstract/highlight constraints, graphical abstract dimensions, unresolved placeholders, and SHA-256 manifest integrity. Raw datasets, checkpoints, TensorRT engines, and per-image predictions are not redistributed in this submission source.

The compact journal source is intentionally limited to manuscript sources, figures, bibliography, and generated tables. It is not itself the evidence archive and does not rerun archive validation during LaTeX compilation.

S15 Minimum reporting checklist for paired corruption studies

For reuse of this protocol, a report should state at least:

- the exact clean materialization and whether corrupted arms share its terminal codec;
- the treatment-level quantization recipe, graph coverage, calibration identities, runtime version, and decoder;
- the atomic image IDs and proof that compared arms consume identical ordered inputs;
- the direct clean-adjusted contrast, not only corrupted treatment means;
- absolute corrupted AP and explicit low-accuracy floor inventories;
- the resampling unit, pairing structure, draw reuse, and uncertainty boundary;
- the finite-grid weighting target and all non-pooled endpoint families;
- sensitivity to split reuse, corruption realization, training and calibration seeds, class universe, and recipe transfer where feasible.